MACHINE LEARNING

Learning from Data

Chapter 1

How observed examples become predictions for new cases

Regression and classification

Regression

How much will the monthly rent be?

A numerical target

Example: 1600

Classification

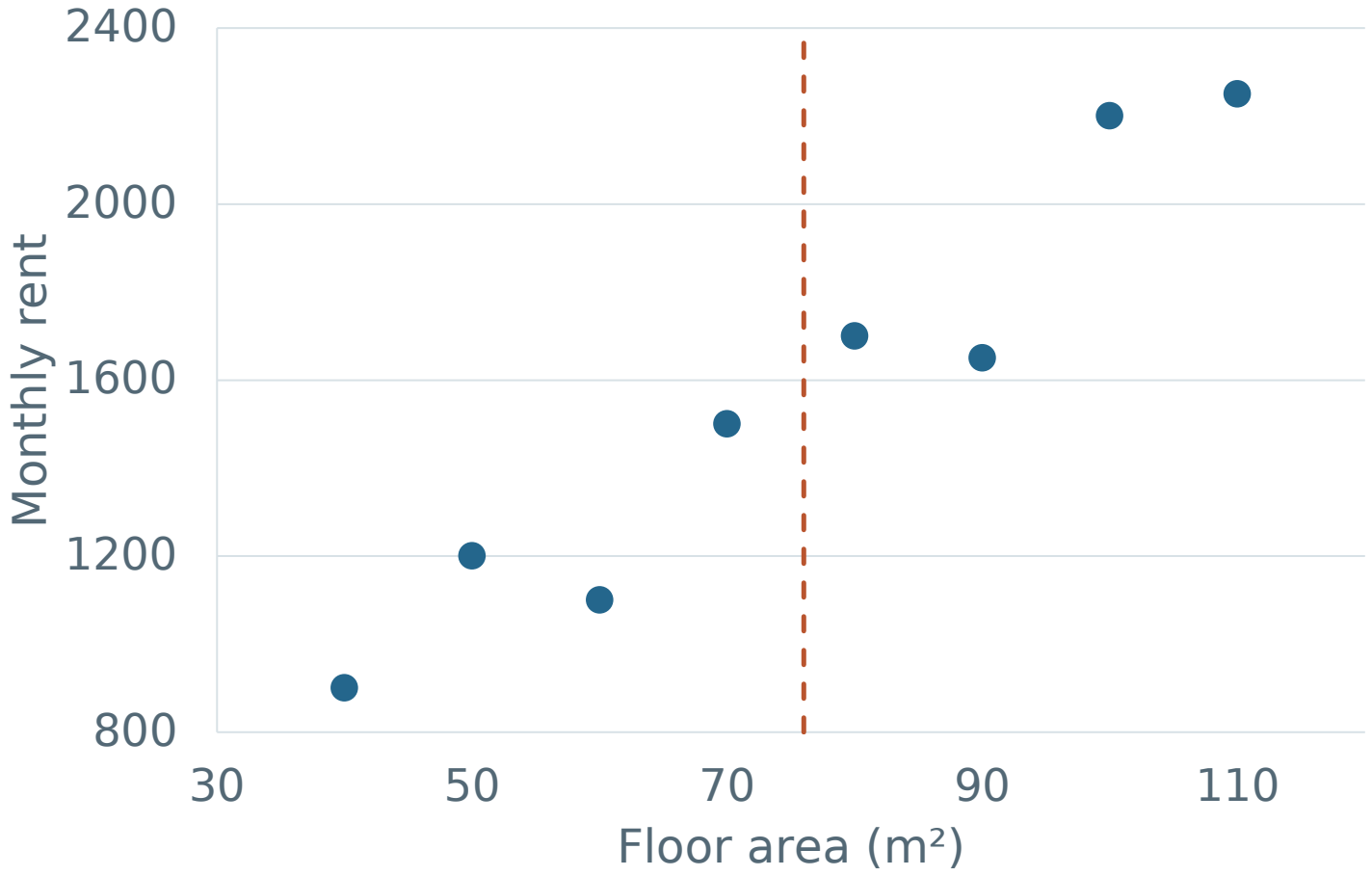
Will the apartment rent within one week?

A categorical target

Within one week / later

Supervised learning uses examples with known targets.

A prediction for a new apartment



76 m²

What rent would you predict?

What connects this apartment
to the observed cases?

Inputs, targets, and observations

Feature (Recorded property)	Value
Floor area	76 m²
Bedrooms	2
Neighborhood	Amir Abad
Target to predict: Monthly Rent	

One instance

An apartment

Input x

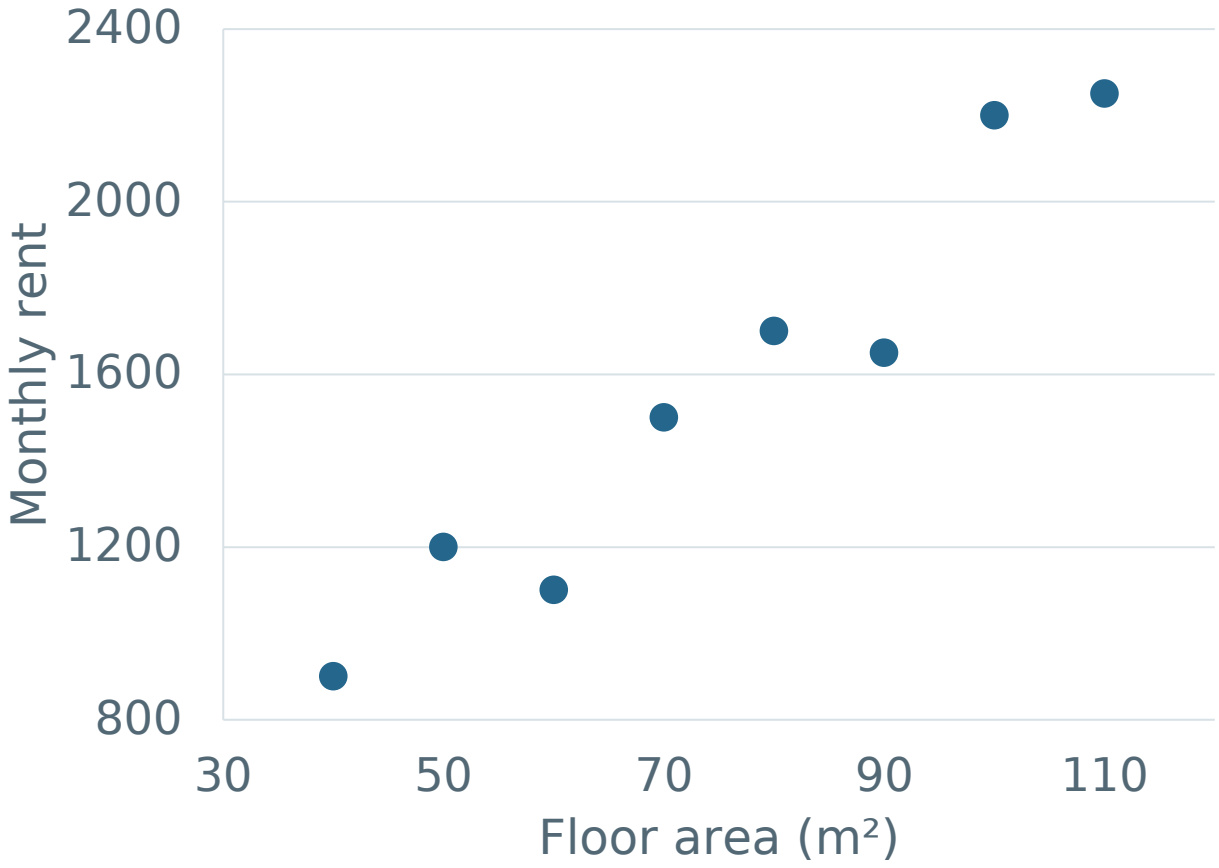
The features we choose to record

Target y

The quantity we want to predict

The training data: Apartments

Area (m²)	Monthly rent
40	900
50	1200
60	1100
70	1500
80	1700
90	1650
100	2200
110	2250



$D = \{(x_1, y_1), \ldots, (x_8, y_8)\}$

A learning algorithm and its model

Training



Prediction



Memorization handles exact matches

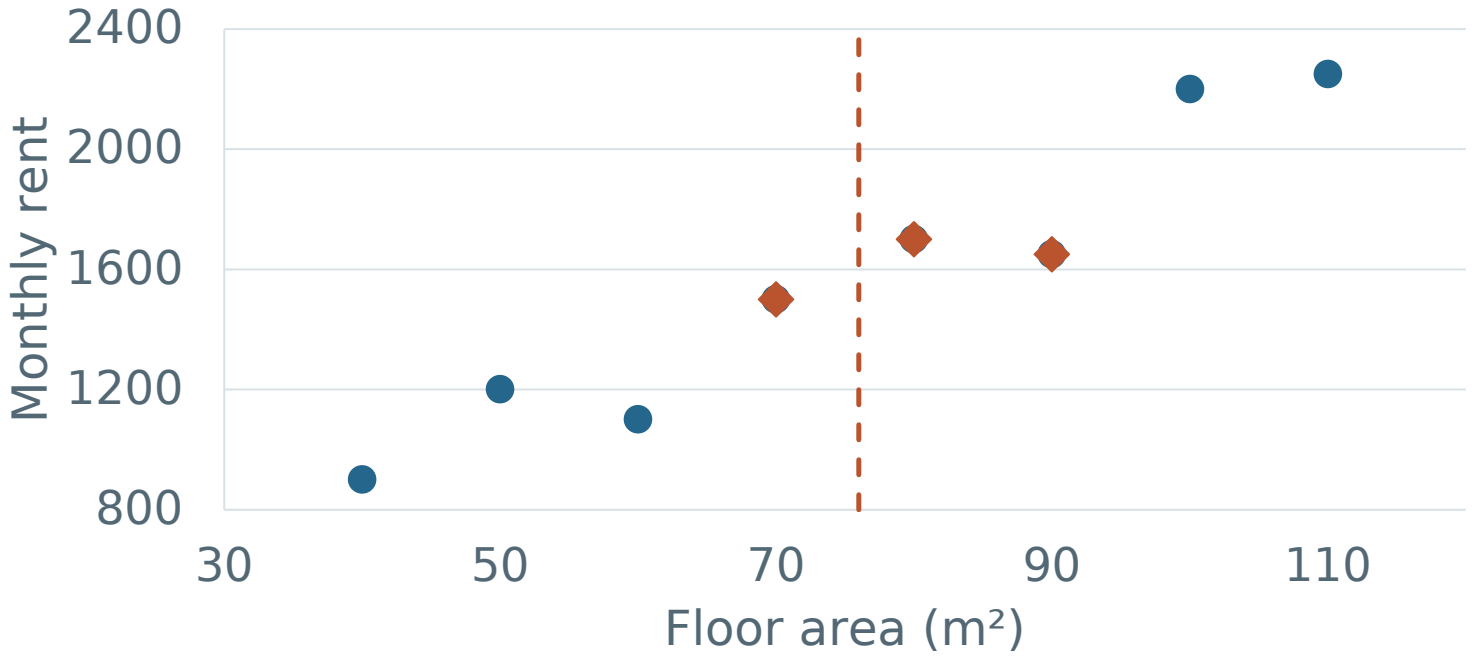
Query area	Lookup result
70 m²	1500
80 m²	1700
76 m²	**No stored answer**

Every training input has an answer

A new input needs a prediction rule

Generalization needs a modeling assumption

Apartments with similar floor areas may have similar rents.



Generalization

Useful predictions for relevant new cases

One-nearest-neighbor prediction

Distance in this example: $d(x, x_i) = |x - x_i|$

Training area	Distance from 76	Observed rent
70	6	1500
80	**4**	**1700**
90	14	1650

Nearest area

80 m²

$$\hat{y} = 1700$$

Averaging several neighbors

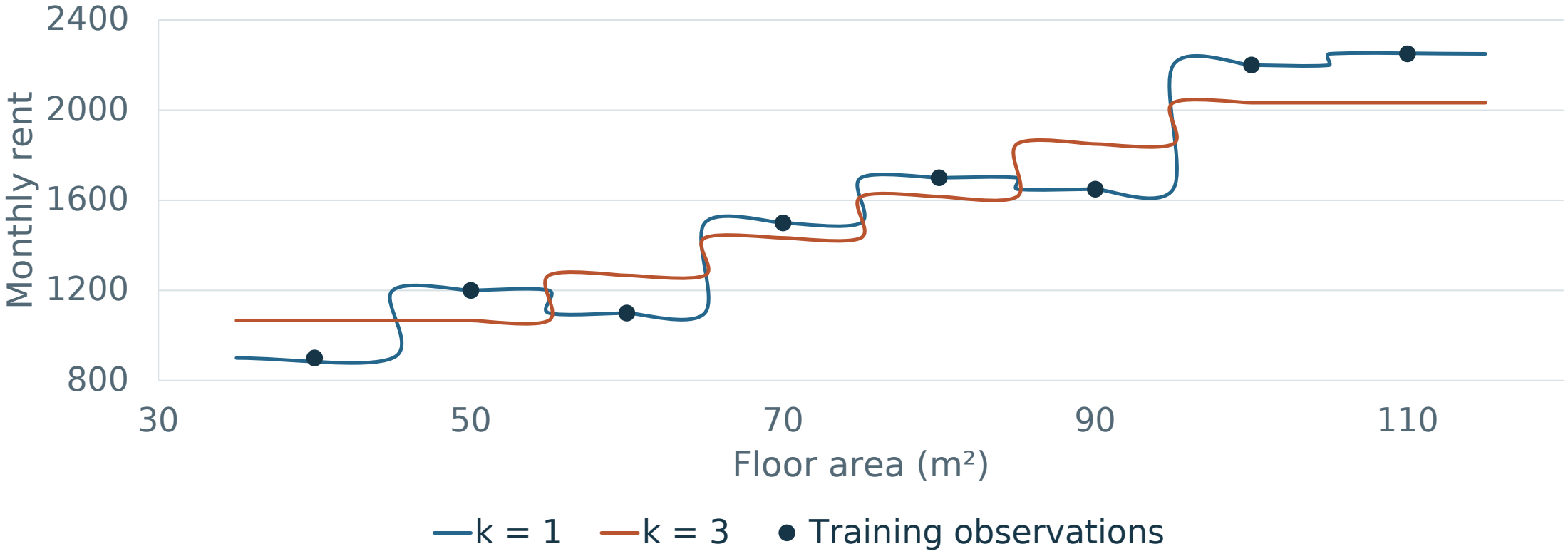
Neighbor order	Area (m²)	Observed rent
1	80	1700
2	70	1500
3	90	1650

k = 3

$$\hat{y} = (1700 + 1500 + 1650) / 3$$

$$\hat{y} \approx 1616.7$$

The fitted rule depends on k



$$\widehat{f}_k(x) = \left(\frac{1}{k}\right)\sum y_i \ \text{, for } i \in N_{\text{k}}(x)$$

Comparing the error on one new case

Area 76 m², observed rent y = 1600

Model	Prediction ŷ	Squared error: (y − ŷ)²
k = 1	1700	10,000
k = 3	**1616.7**	**277.8**

Averaging the neighbors gives the smaller error on this case.

Mean squared error

$$MSE = \frac{\sum_{1}^{m}(y_i - \hat{y}_i)^2}{m}$$

y_i	Observed target
$\hat{y}_i$	Model prediction
m	Number of evaluated cases

Training error can reward memorization

Target for every training example will be predicted accurately

MSE = 0

Nothing can be said for new cases

The prediction task concerns relevant new cases.

Validation data compare candidate models

Area (m²)	Observed rent	k = 1 prediction	Squared error	k = 3 prediction	Squared error
54	1100	1200	10,000	1066.7	1111.1
76	1600	1700	10,000	1616.7	277.8
96	1950	2200	62500	2033.3	6944.4

Squared errors use unrounded predictions.

Choosing k with validation error

k = 1

(10,000 + 10,000 + 62,500) / 3

MSE = 27,500

k = 3

(1111.1 + 277.8 + 6944.4) / 3

MSE ≈ 2777.8

Selected on this validation set

Validation results influence model selection

Validation result

**We chose k = 3 because
it had lower validation MSE.**

A final assessment needs data that played no role in those choices.

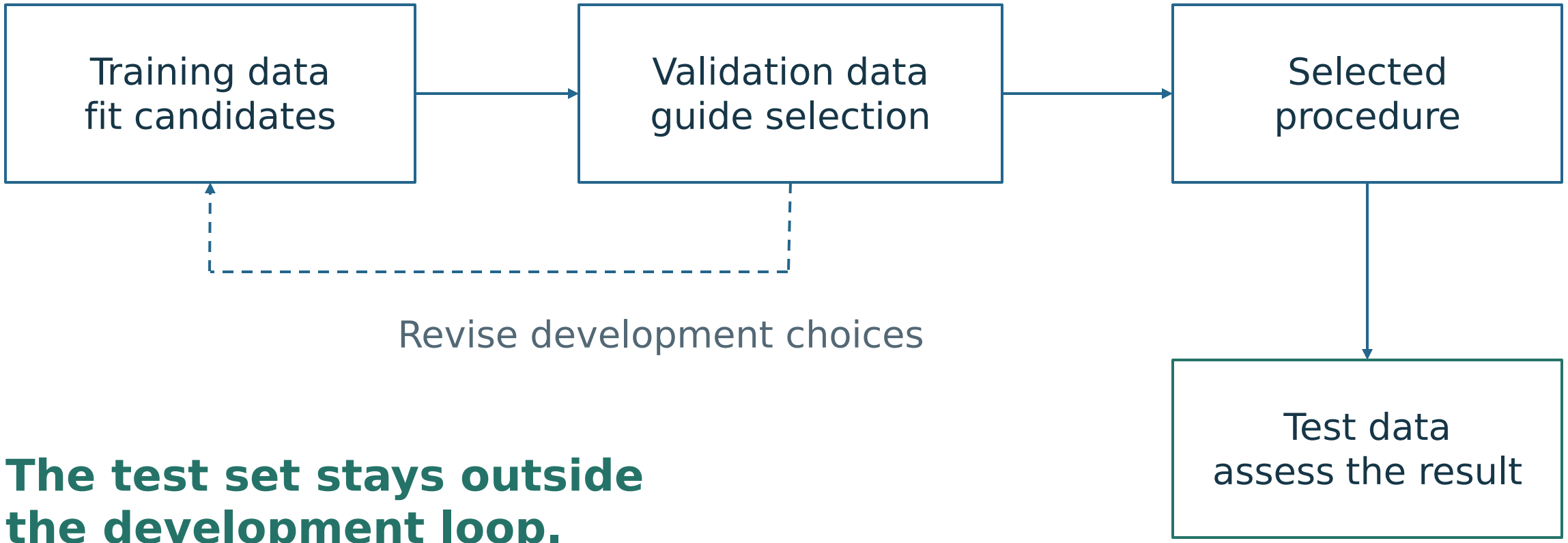
A separate test set

Area (m²)	Observed rent	Selected k = 3 prediction
64	1300	1266.7
84	1650	1616.7
104	2100	2033.3

Test MSE ≈ 2222.2

A finite-sample estimate of performance on new cases

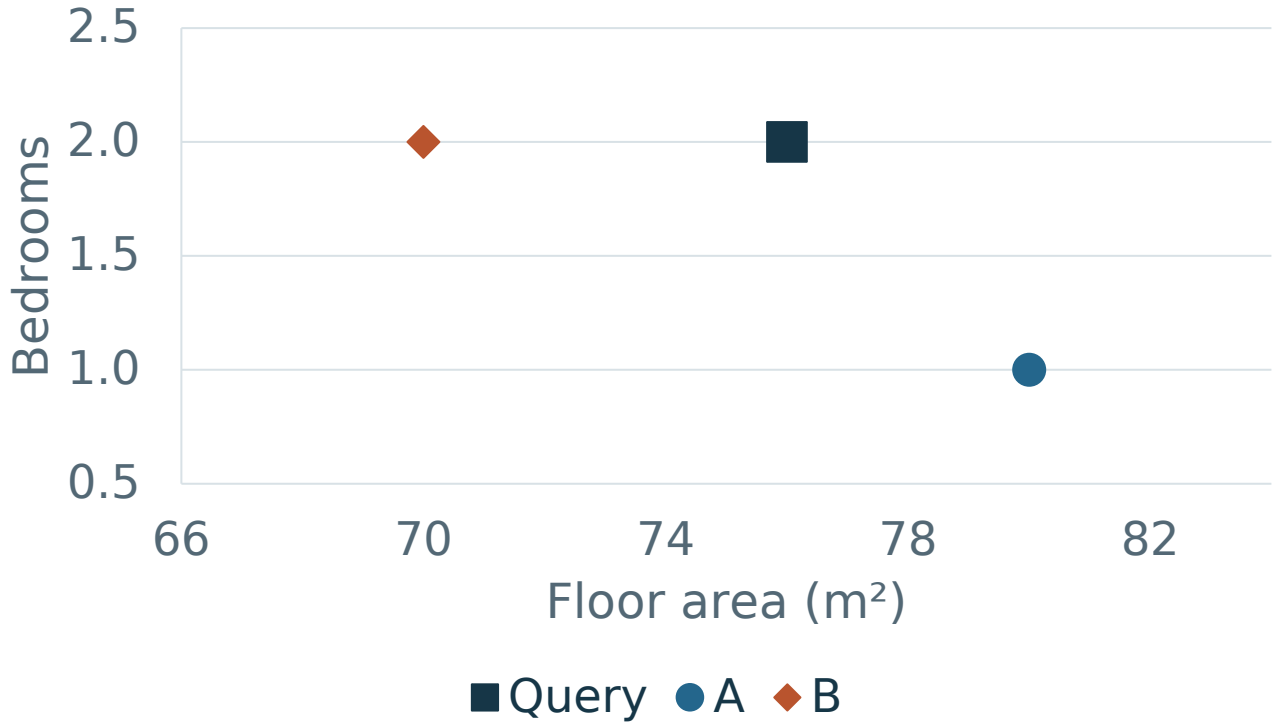
Training, validation, and test



The test set stays outside the development loop.

Feature scales define closeness

Query x = (76 m², 2 bedrooms)



Case	Squared distance
A: (80, 1)	**$4^2 + 1^2 = 17$**
B: (70, 2)	$6^2 + 0^2 = 36$

A is closer under this comparison.

Rescaling a feature can change the neighbor

Measure area in units of 10 m².

Candidate	Original squared distance	Rescaled squared distance
A	$4^2 + 1^2 = 17$	$0.4^2 + 1^2 = 1.16$
B	$\mathbf{6^2 + 0^2 = 36}$	$\mathbf{0.6^2 + 0^2 = 0.36}$

B is now closer.

Feature scales express relative importance in this distance rule.

Nearest neighbors can also classify

Same eight apartments. New target: rented within one week (Quick) or Later.

Area (m²)	Class target
40	Later
50	Later
60	**Quick**
70	**Quick**

Area (m²)	Class target
80	**Quick**
90	Later
100	Later
110	Later

Use floor area to predict a class label instead of a rent amount.

The class targets are illustrative observed rental outcomes.

Classification uses a neighbor vote

New apartment: 76 m². Use the same distance rule and k = 3.

Neighbor	Area (m²)	Distance	Class
1	80	4	Quick
2	70	6	Quick
3	90	14	Later

2 Quick, 1 Later

Predict Quick

Regression averages numerical targets. Classification votes over class labels.

Select k for the classification task using its own validation data.

Positive and negative classes

For this evaluation, choose Quick as the positive class.

Class	Meaning	Possible code
Positive (+)	Quick: rented within one week	1
Negative (−)	Later: rented after one week	0

Actual class What happened after the apartment was listed.

Predicted class What the fitted model predicted.

Four possible prediction outcomes

Positive = Quick. Negative = Later.

Outcome	Prediction	Actual outcome
True positive (TP)	Quick	Quick
False positive (FP)	Quick	Later
True negative (TN)	Later	Later
False negative (FN)	Later	Quick

True / false describes correctness. Positive / negative identifies the prediction.

A confusion matrix counts four outcomes

Illustrative held-out evaluation: 200 apartments. Positive class = Quick.

Actual class	Predicted Quick (+)	Predicted Later (−)	Total
Quick (+)	TP = 30	FN = 20	50
Later (−)	FP = 10	TN = 140	150
Total	40	160	200

Accuracy = (TP + TN) / 200 = 170/200 = .85

Error rate = (FP + FN) / 200 = 30/200 = .15

Consequences of classification errors

Suppose an agency shortlists apartments predicted to rent within a week.

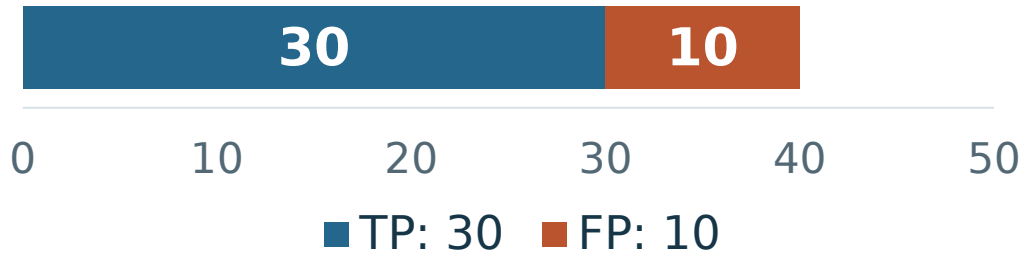
False positive	10 slower apartments entered the Quick shortlist.
False negative	20 apartments that rented quickly were missed.

Precision: shortlist reliability. Recall: coverage of Quick apartments.

Precision and recall

Precision

Of the predicted Quick apartments, how many actually rented quickly?



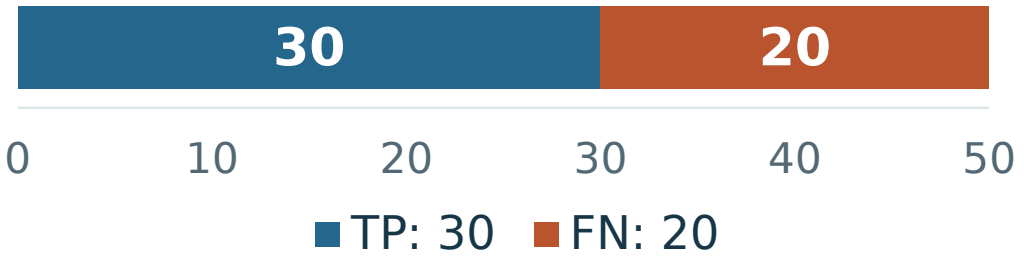
Precision = TP / (TP + FP)

= 30/40 = .75

75% of the shortlist was correct.

Recall

Of all actually Quick apartments, how many did the model identify?



Recall = TP / (TP + FN)

= 30/50 = .60

60% of Quick apartments were found.

F1 combines precision and recall

F1 is the harmonic mean of precision P and recall R.

$$F1 = 2PR / (P + R)$$

$$= 2TP / (2TP + FP + FN)$$

$$= 60 / (60 + 10 + 20) = 2/3 \approx .667$$

High F1 requires both good precision and good recall.

True negatives do not enter this formula. State a convention for zero denominators.

Matthews correlation coefficient (MCC)

A correlation between the actual and predicted class labels.

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}$$

$MCC = 4000 / \sqrt{48{,}000{,}000} \approx .5774$

+1: perfect 0: no association −1: reversed

Class-name swaps leave MCC unchanged. Define how zero denominators are handled.

Changing the positive class

Same predictions. Now evaluate Later as the positive class.

Positive class	Precision	Recall	F1	MCC
Quick	30/40 = .750	30/50 = .600	2/3 ≈ .667	.5774
Later	140/160 = .875	140/150 ≈ .933	28/31 ≈ .903	.5774

After the swap: TP = 140, FP = 20, FN = 10, TN = 30.

Higher scores for Later describe a different class, with the same predictions.

Macro averaging: equal class importance

If neither class has priority, compute each class’s score and average the two.

Measure	Quick positive	Later positive	Macro average
Precision	.750	.875	(.750+.875)/2 = .8125
Recall	.600	.9333	(.600+.9333)/2 ≈ .7667
F1	.6667	.9032	(.6667+.9032)/2 ≈ .7849

Macro F1 = $(F1_{Quick} + F1_{Later}) / 2$

Average the class F1 scores, not the already-averaged precision and recall.

Macro, weighted, and micro averages

The evaluation set contains 50 Quick and 150 Later apartments.

Average	What gets weight	F1 here
Macro	Each class equally: ½ and ½	.7849
Support-weighted	Class frequencies: ¼ and ¾	.8441
Micro	Pool the class confusion counts first	.8500

For this single-label task, micro F1 equals accuracy.

State the averaging rule. Macro expresses equal class importance.

Discussion: an always-Later classifier

What happens if every apartment is predicted Later?

Rule	TP	FP	FN	TN
Given classifier	30	10	20	140
Always Later	0	0	50	150

Calculate accuracy, Quick recall, and Quick F1 for both rules.

Which measures reveal that the constant rule finds no Quick apartments?

Reporting classification performance

Class of interest	State which class is positive and why.
Per-class behavior	Report precision and recall for the classes that matter.
Overall summary	Name the averaging rule, and consider MCC or accuracy.
Final assessment	Use test data after the modeling choices are complete.